

Justify your answers and show your work. Unless indicated otherwise, each problem is worth 10 points.

1. $\int \cos^3 2x \, dx$

2. $\int \sin^2 x \, dx$

3. $\int \sin^3 x \cos x \, dx$

4. $\int \frac{dx}{\sqrt{25-x^2}}$

5. $\int_0^7 \sqrt{49-x^2} \, dx$. (Hint: $\int \cos^2 x \, dx = \frac{x}{2} + \frac{\sin(2x)}{4} + C$)

6. $\int \frac{x^2}{(\sqrt{25-x^2})^3} \, dx$ (Hint: $\int \frac{\sin^2 \theta}{\cos^2 \theta} \, d\theta = \int \tan^2 \theta \, d\theta = \tan \theta + \theta + C$) [5pts]

7. $\int \frac{x}{x^2 + x - 12} dx$ [15 pts]

8. $\int_0^{\ln 2} 4xe^x dx$

9. $\int_1^{\infty} 6x^{-2} dx$

10. $\int_2^{\infty} \frac{17}{x \ln x} dx$